

PROJECT-1

Multi Tier Website Deployment

The image displays two screenshots from the AWS Management Console, illustrating the deployment of a multi-tier website.

Top Screenshot: AWS Management Console - EC2 Instances

The top screenshot shows the AWS Management Console interface for the 'ap-south-1' region. The 'Instances' page is active, displaying a list of EC2 instances. A single instance, 'Project-Instance' (ID: i-09b71c39e4c5bbb03), is shown in the 'Running' state. The instance type is 't2.micro'. The 'Status check' is 'Initializing'. The 'Alarm status' is 'View alarms +'. The 'Launch instances' button is visible in the top right corner.

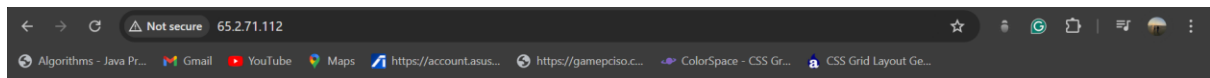
Bottom Screenshot: CloudShell Terminal

The bottom screenshot shows a CloudShell terminal session connected to the EC2 instance. The terminal output indicates that no containers need to be restarted and that no VM guests are running outdated hypervisor (qemu) binaries. The user 'ubuntu' is logged in as 'ip-172-31-2-124'. The terminal shows the following commands and output:

```
ubuntu@ip-172-31-2-124:~$ cd var/www/html
-bash: cd: var/www/html: No such file or directory
ubuntu@ip-172-31-2-124:~$ cd /
ubuntu@ip-172-31-2-124:/$ cd var/www/ht
-bash: cd: var/www/ht: No such file or directory
ubuntu@ip-172-31-2-124:/$ cd var/www/html/
ubuntu@ip-172-31-2-124:/var/www/html$ ls
index.html
ubuntu@ip-172-31-2-124:/var/www/html$ sudo rm index.html
ubuntu@ip-172-31-2-124:/var/www/html$ ls
ubuntu@ip-172-31-2-124:/var/www/html$ sudo nano index.php
ubuntu@ip-172-31-2-124:/var/www/html$ ls
index.php
ubuntu@ip-172-31-2-124:/var/www/html$
```

The terminal also shows the instance's public and private IP addresses:

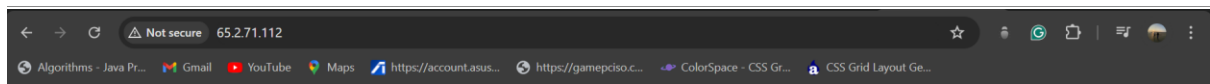
```
i-09b71c39e4c5bbb03 (Project-Instance)
PublicIPs: 65.2.71.112 PrivateIPs: 172.31.2.124
```



Name:

Email:

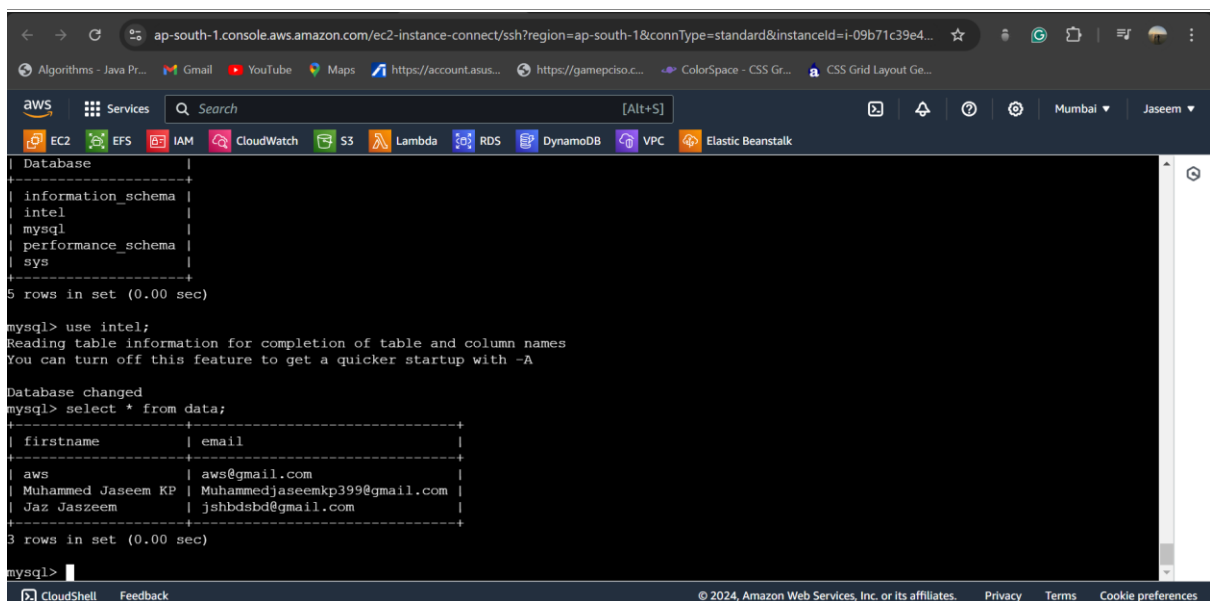
```
connect_error) { die("Connection failed: " . $conn->connect_error); } if(isset($_POST['firstname']) && isset($_POST['email'])) { $sql = "INSERT INTO data (firstname,email) VALUES ('".$_POST['firstname']."', '".$_POST['email']."')"; if ($conn->query($sql) === TRUE) { echo "New record created successfully"; } else { echo "Error: " . $sql . " " . $conn->error; } $conn->close(); } ?>
```



Name:

Email:

New record created successfully



ap-south-1.console.aws.amazon.com/ec2/home?region=ap-south-1#CreateTargetGroup:

Target group name

Project-TG

A maximum of 32 alphanumeric characters including hyphens are allowed, but the name must not begin or end with a hyphen.

Protocol : Port

Choose a protocol for your target group that corresponds to the Load Balancer type that will route traffic to it. Some protocols now include anomaly detection for the targets and you can set mitigation options once your target group is created. This choice cannot be changed after creation

HTTP 80

1-65535

IP address type

Only targets with the indicated IP address type can be registered to this target group.

☒ IPv4

Each instance has a default network interface (eth0) that is assigned the primary private IPv4 address. The instance's primary private IPv4 address is the one that will be applied to the target.

☐ IPv6

Each instance you register must have an assigned primary IPv6 address. This is configured on the

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ap-south-1.console.aws.amazon.com/ec2/home?region=ap-south-1#TargetGroup.targetGroupArn=arn:aws:elasticloadbalancing:ap-south-1:361769599900:targetgroup/Project-TG/d3cb78bef8800c5a

Project-TG

Details

arn:aws:elasticloadbalancing:ap-south-1:361769599900:targetgroup/Project-TG/d3cb78bef8800c5a

Target type	Protocol : Port	Protocol version	VPC
Instance	HTTP: 80	HTTP1	vpc-047e0ec5b5652a2bc
IP address type	Load balancer		
IPv4	None associated		

1	0	0	1	0	0
Total targets	Healthy	Unhealthy	Unused	Initial	Draining
	0 Anomalous				

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ap-south-1.console.aws.amazon.com/ec2/home?region=ap-south-1#CreateALBWizard:

How Application Load Balancers work

Basic configuration

Load balancer name

Name must be unique within your AWS account and can't be changed after the load balancer is created.

Project-LB

A maximum of 32 alphanumeric characters including hyphens are allowed, but the name must not begin or end with a hyphen.

Scheme

Scheme can't be changed after the load balancer is created.

☒ Internet-facing

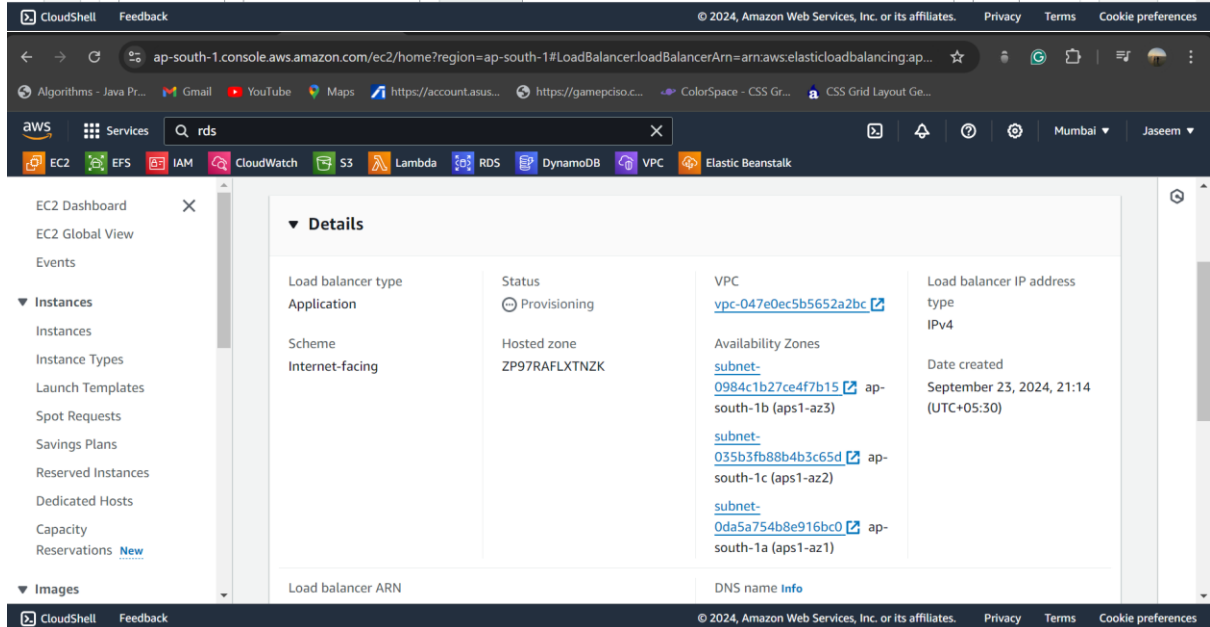
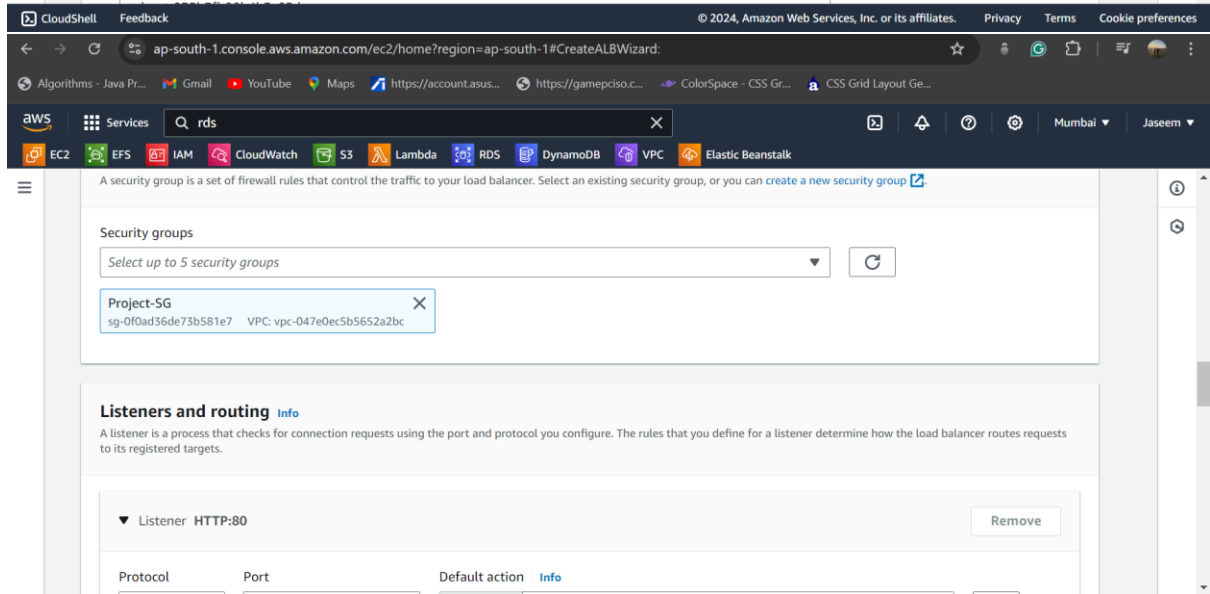
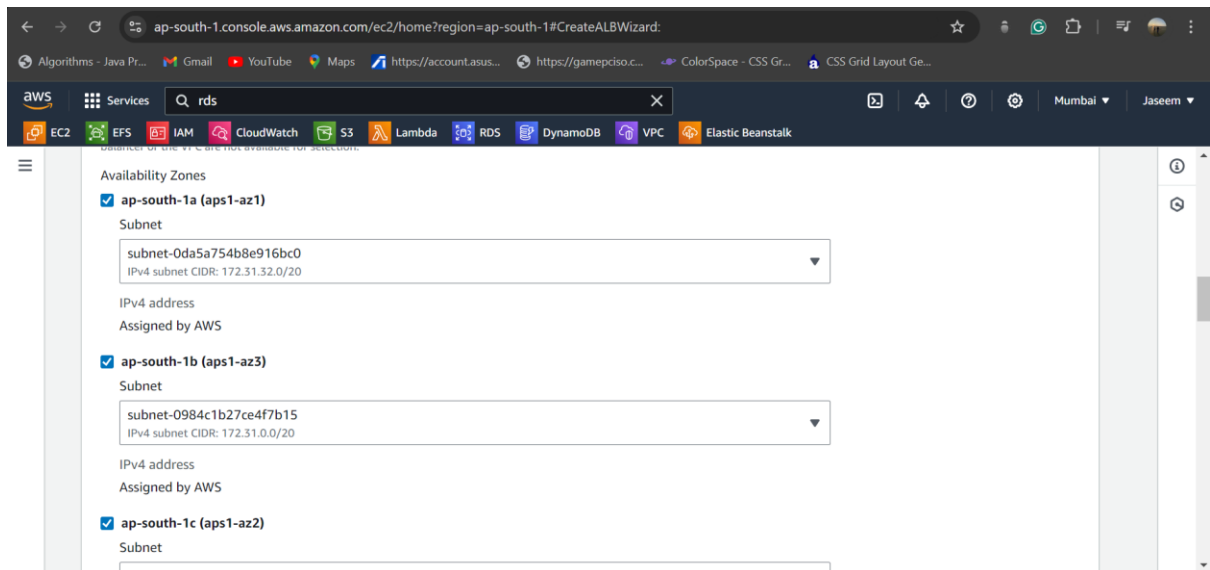
An internet-facing load balancer routes requests from clients over the internet to targets. Requires a public subnet. [Learn more](#)

☐ Internal

An internal load balancer routes requests from clients to targets using private IP addresses. Compatible with the IPv4 and Dualstack IP address types.

Load balancer IP address type

Select the front-end IP address type to assign to the load balancer. The VPC and subnets mapped to this load balancer must include the selected IP address types. Public IPv4 addresses have



ap-south-1.console.aws.amazon.com/ec2/home?region=ap-south-1#TargetGroup.targetGroupArn=arn:aws:elasticloadbalancing:ap-s...

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EC2 EFS IAM CloudWatch S3 Lambda RDS DynamoDB VPC Elastic Beanstalk

EC2 Dashboard X

EC2 Global View

Events

▼ Instances

Instances

Instance Types

Launch Templates

Spot Requests

Savings Plans

Reserved Instances

Dedicated Hosts

Capacity

Reservations New

▼ Images

Distribution of targets by Availability Zone (AZ)

Select values in this table to see corresponding filters applied to the Registered targets table below.

Targets Monitoring Health checks Attributes Tags

Registered targets (1) Info Anomaly mitigation: Not applicable Deregister Register targets

Target groups route requests to individual registered targets using the protocol and port number specified. Health checks are performed on all registered targets according to the target group's health check settings. Anomaly detection is automatically applied to HTTP/HTTPS target groups with at least 3 healthy targets.

Filter targets

☐	Instance ID	Name	Port	Zone	Health status	Health
☐	i-09b71c39e4c5bbb03	Project-Instance	80	ap-south-1b	Healthy	-

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Not secure project-lb-1271826230.ap-south-1.elb.amazonaws.com

FIFA 22: End of an E... Gmail YouTube Maps ColorSpace - CSS Gr... CSS Grid Layout Ge... ASUS Software Port... MyASUS Software ... McAfee LiveSafe

Name:

Email:

Submit

ap-south-1.console.aws.amazon.com/ec2/home?region=ap-south-1#CreateImage.instanceId=i-09b71c39e4c5bbb03

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Create image Info

An image (also referred to as an AMI) defines the programs and settings that are applied when you launch an EC2 instance. You can create an image from the configuration of an existing instance.

Instance ID

i-09b71c39e4c5bbb03 (Project-Instance)

Image name

Project-AMI

Maximum 127 characters. Can't be modified after creation.

Image description - optional

Image description

Maximum 255 characters

☒ Reboot instance

When selected, Amazon EC2 reboots the instance so that data is at rest when snapshots of the attached volumes are taken. This ensures data consistency.

Instance volumes

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ap-south-1.console.aws.amazon.com/ec2/home?region=ap-south-1#images:visibility=owned-by-me

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Launch Templates
Spot Requests
Savings Plans
Reserved Instances
Dedicated Hosts
Capacity
Reservations **New**

▼ Images
AMIs
AMI Catalog

▼ Elastic Block Store
Volumes
Snapshots
Lifecycle Manager

Amazon Machine Images (AMIs) (1) Info

Recycle Bin EC2 Image Builder Actions Launch instance from AMI

Owned by me Find AMI by attribute or tag

☐	Name	AMI name	AMI ID	Source
☐		Project-AMI	ami-0fc161127d6075c60	361769599900/Project-AMI

Select an AMI

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ap-south-1.console.aws.amazon.com/ec2/home?region=ap-south-1#LaunchTemplates:

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EC2 Dashboard
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Events

▼ Instances
Instances
Instance Types
Launch Templates
Spot Requests
Savings Plans
Reserved Instances
Dedicated Hosts
Capacity
Reservations **New**

▼ Images

Launch Templates (1) Info

Search

☐	Launch Template ID	Launch Template Name	Default Version	Latest Version	Create Time
☐	lt-017d16d9b827a7a69	Project-Template	1	1	2024-09-23T

Select a launch template

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ap-south-1.console.aws.amazon.com/ec2/home?region=ap-south-1#AutoScalingGroups:

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EC2 > Auto Scaling groups

Auto Scaling groups (1) Info

Launch configurations Launch templates Actions Create Auto Scaling group

Search your Auto Scaling groups

☐	Name	Launch template/configuration	Instances	Status	Desired capacity	Min
☐	Project-ASG	Project-Template Version Default	0	Updating capacity...	2	1

0 Auto Scaling groups selected

ap-south-1.console.aws.amazon.com/ec2/home?region=ap-south-1#AutoScalingGroups?id=Project-ASG;view=activity

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EC2 > Auto Scaling groups

Auto Scaling groups (1/1) Info Launch configurations Launch templates Actions Create Auto Scaling group

Search your Auto Scaling groups

Name	Launch template/configuration	Instances	Status	Desired capacity
Project-ASG	Project-Template Version Default	0	Updating capacity...	2

Auto Scaling group: Project-ASG

Successfull

At 2024-09-23T15:59:42Z an instance was started in response to a difference between desired and actual capacity, increasing the capacity from 1 to 2.

September 23, 09:23 PM +05

Successfull

At 2024-09-23T15:59:28Z a user request created an AutoScalingGroup changing the desired capacity from 0 to 2. At 2024-09-23T15:59:37Z an instance was started in response to a difference between

2024 September

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ap-south-1.console.aws.amazon.com/ec2/home?region=ap-south-1#instances:v=3;\$case=tags:true%5C,client:false;\$regex=tags:false%5C,c...

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EC2 Dashboard EC2 Global View Events

Instances

Instances Instance Types Launch Templates Spot Requests Savings Plans Reserved Instances Dedicated Hosts Capacity Reservations New

Images

Instances (5) Info Last updated less than a minute ago Connect Instance state Actions Launch instances

Find Instance by attribute or tag (case-sensitive) All states

	Name	Instance ID	Instance state	Instance type	Status check	Alarm status
☐		i-01000220ad36da6e8	Running	t2.micro	Initializing	View alarms +
☐		i-02023c26c0e913c7e	Running	t2.micro	Initializing	View alarms +
☐	Project-Instance	i-0487f700964b4544d	Terminated	t2.micro	-	View alarms +
☐	Project-Instance	i-06602a50883aae8e7	Terminated	t2.micro	-	View alarms +

Select an instance

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ap-south-1.console.aws.amazon.com/ec2/home?region=ap-south-1#instances:v=3;\$case=tags:true%5Cclient:false;\$regex=tags:false%5C,c...

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EC2 Dashboard EC2 Global View Events

Instances Instances Instance Types Launch Templates Spot Requests Savings Plans Reserved Instances Dedicated Hosts Capacity Reservations New

Images

Instances (1/5) Info Last updated less than a minute ago Connect Instance state Actions Launch instances

Find Instance by attribute or tag (case-sensitive) All states < 1 >

	Name	Instance ID	Instance state	Instance type	Status check	Alarm status
☒		i-01000220ad36da6e8	Running	t2.micro	Initializing	View alarms +
☐		i-02023c26c0e913c7e	Running	t2.micro	Initializing	View alarms +
☐	Project-Instance	i-0487f700964b4544d	Terminated	t2.micro	-	View alarms +
☐	Project-Instance	i-06607a50883aae8e7	Terminated	t2.micro	-	View alarms +

i-01000220ad36da6e8

Instance summary Info

Instance ID i-01000220ad36da6e8

IPv6 address

Public IPv4 address copied 15.206.69.189 | open address

Private IPv4 addresses 172.31.33.156

Instance state

Public IPv4 DNS

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Not secure 15.206.69.189

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